

The Rhythm of Reason: Decrypting the "Black Box" of Fan Voting and Optimizing the *Dancing with the Stars* Ecosystem

Summary

For thirty-four seasons, *Dancing with the Stars* has captivated audiences by blending ballroom excellence with celebrity charisma. However, the mechanism combining professional judge scores and public fan votes remains a "Black Box," historically sparking controversy when popular contestants survive despite poor technical performance. This competition seeks to decode these dynamics and propose a fairer aggregation model.

Decoding the Invisible Hand: First, to overcome the challenge of unobservable fan data, we developed an **Inverse Bayesian Reconstruction Model**. By treating historical elimination outcomes as strict constraints on a Dirichlet distribution, we employed Monte Carlo simulations to infer the latent voting shares for every week. The model achieved a **98.3%** historical reproduction rate and revealed a "Bubble Effect," where voting certainty is highest for at-risk contestants.

Analyzing the Popularity Machine: Second, utilizing a **Linear Mixed-Effects Model**, we quantified the drivers of success. We discovered a fundamental divergence: judges evaluate a multi-factor technical matrix (Skill, Partner, Age), whereas fans overwhelmingly vote for pre-existing **Base Popularity** regardless of weekly performance. This finding explains the structural vulnerability of linear aggregation rules to "popularity monopolies."

Simulating Parallel Universes: Third, we established a **Dual-Track Counterfactual Framework** to compare the show's two historical systems: Rank-based (The "Shield") and Percent-based (The "Megaphone"). Simulations confirmed that while the Rank system prevents monopoly, it suppresses fan enthusiasm. Conversely, the Percent system suffers from "Linear Vulnerability," allowing anomalies like Season 27's Bobby Bones to become mathematically invincible.

Optimizing the Future: Finally, we propose the **Adaptive Logistic Meritocracy System (ALMS)**. This novel mechanism introduces a Sigmoid saturation curve to impose a "soft ceiling" on extreme popularity and dynamically adjusts sensitivity based on competition intensity. Stress-testing demonstrates that ALMS achieves a Pareto improvement: increasing technical fairness by **13.3%** while retaining **95.4%** of fan satisfaction. We recommend ALMS, alongside the "Judges' Save," as the optimal strategy for the show's post-game era.

Keywords: Inverse Bayesian Inference; Monte Carlo Simulation; Linear Mixed-Effects Model; Social Choice Theory; Adaptive System; Dancing with the Stars

Contents

1	Introduction	2
1.1	Background	2
1.2	Restatement of the Problem	2
1.3	Our Work	3
2	Assumptions	4
3	Notations	4
4	Model I: Bayesian Estimation of Fan Voting Distribution	5
4.1	Problem Formulation	5
4.2	Mathematical Model Establishment	5
4.3	Algorithm: Iterative Inverse Estimator	6
4.4	Validation and Results	7
5	Model II: Analysis of Celebrity Characteristics	9
5.1	Feature Interpretation and Engineering	9
5.2	Core Model Construction	10
5.3	Validation Strategy and Metrics	10
5.4	Results and Discussion	11
6	Model III: Dynamic Comparison and Optimization of the Combination Systems	13
6.1	Problem Formulation: The Dual-Track Simulation Framework	13
6.2	Analysis Results Comparison	14
6.3	Mechanism Analysis	15
6.4	Policy Optimization: A Multi-Objective Evaluation	16
7	Model IV: The Adaptive Logistic Meritocracy System	17
7.1	Problem Formulation and Motivation	17
7.2	Mathematical Model Establishment	18
7.3	Simulation and Validation	20
7.4	Comparative Evaluation	20
7.5	Conclusion	21
8	Memorandum	22
9	Model Evaluation and Promotion	23
9.1	Model Evaluation	23
9.2	Future Work	23
10	Conclusions	23

1 Introduction

1.1 Background

"Dancing with the Stars" (DWTS) is a hit reality competition pairing celebrities with professional dancers. Each week, pairs are evaluated through a hybrid mechanism: judge scores from experts and fan votes from the audience. These two inputs are combined to determine which couple is eliminated, ultimately crowning a champion in the finale.



Figure 1: The Post of the Show

Over 34 seasons, the show has utilized different aggregation methods—primarily Rank-based and Percent-based systems—to weigh these inputs. However, the clash between dance skill and star power has led to notable controversies, where popular contestants advanced despite low technical scores. This study analyzes these historical voting dynamics to quantify fan influence and proposes an optimized framework to better balance competitive fairness with viewer engagement.

1.2 Restatement of the Problem

The central objective of this research is to decode the decision-making dynamics within "Dancing with the Stars," specifically how professional evaluations interact with public preference to determine contest results. Given the unavailability of raw audience voting data, we must infer these hidden variables to rigorously assess the fairness of the competition. The specific objectives are defined as follows:

- **Task 1: Estimation of Fan Voting Data.** We are required to build a statistical model to infer the undisclosed fan votes for each couple in each week. This involves verifying that our inferred data, when integrated with judge scores, accurately replicates the historical elimination patterns, while also providing a metric for the certainty of these inferences.
- **Task 2: Evaluation of Aggregation Methodologies.** We need to check the historical performance of both Rank-based and Percentage-based voting systems using our fan vote estimates. This includes determining which system excessively caters to fan votes, simulating

whether "controversial" outcomes could have been avoided under different rules, and evaluating the efficacy of the "Judges' Choice" intervention.

- **Task 3: Impact Analysis of Celebrity Characteristics.** We must quantify the influence of auxiliary variables (such as the assigned professional partner and celebrity demographics) on the competition's final standings. A key aspect is to distinguish how these attributes differentially affect experts versus the audience.
- **Task 4: Policy Optimization and Strategic Recommendation.** We need to propose a novel system that optimizes the trade-off between technical guarantee and viewer engagement. Furthermore, we are tasked with compiling our insights into a concise memo for the producers, offering evidence-based guidelines for future seasons.

1.3 Our Work

In this paper, we first conducted rigorous preprocessing on the provided data. We then established distinct yet interconnected mathematical models tailored for each specific task. These models successfully addressed the core requirements of each problem. Furthermore, we integrated our independent insights and additional reflections into each model to enhance the depth of our analysis. The overall workflow of our methodology is illustrated in Figure 2.

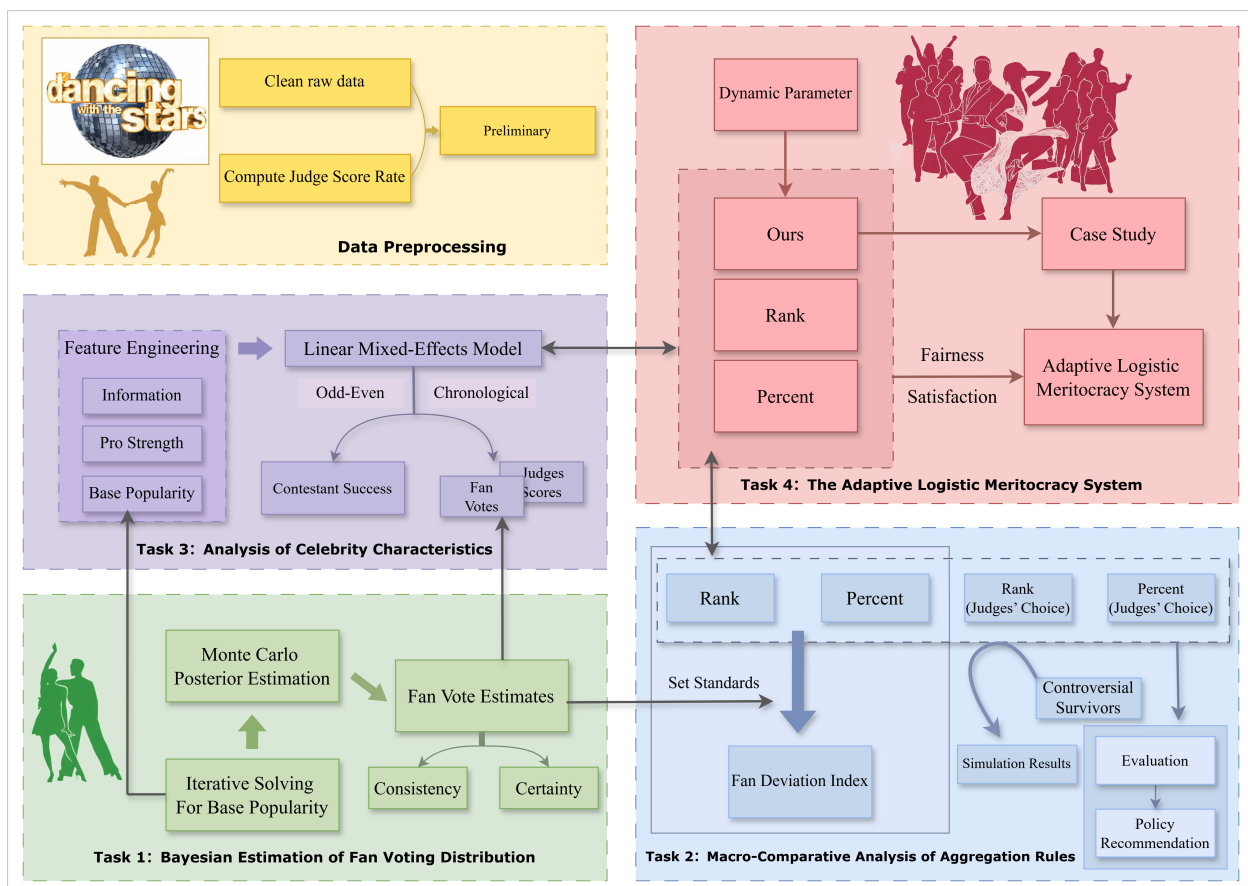


Figure 2: Our Work

2 Assumptions

Model I: Fan Voting Reconstruction

Assumption 1: Performance-Popularity Mixture. The expected fan vote share is a linear combination of a time-invariant *Base Popularity* and the weekly *Judge Performance*.

Assumption 2: Rational Judges' Save. We assume judges consistently save the couple with the higher technical score when deciding between the bottom two.

Assumption 3: Dirichlet Distribution. Fan vote shares are modeled using a Dirichlet distribution to satisfy the simplex constraint ($\sum x_i = 1$).

Assumption 4: Constant Vote Volume. We assume a constant total weekly vote volume ($V_{total} = 14$ million) to normalize cross-season comparisons.

Model II: Factor Analysis

Assumption 5: Partnership Stability. Celebrity-professional pairings are assumed constant; temporary replacements are ignored.

Assumption 6: Sufficient Proxy. The estimated *Base Popularity* fully captures the celebrity's pre-existing fan base variance.

Assumption 7: Linearity of Effects. The impact of characteristics (e.g., age, industry) on placement is linear and additive within the mixed-effects model.

Model III: Policy Simulation

Assumption 8: Rule Independence. Contestant performance and popularity are intrinsic and independent of the aggregation rule employed.

Assumption 9: Metric Comparability. The Manhattan distance between the actual ranking and the pure fan ranking effectively quantifies "Fan-Friendliness".

Model IV: Adaptive Logistic Meritocracy System

Assumption 10: Diminishing Marginal Utility. The effective influence of fan votes follows a logistic growth pattern, saturating at extreme values to prevent monopoly.

Assumption 11: Dynamic Sensitivity. The optimal discrimination parameter K is inversely proportional to the vote distribution's standard deviation.

3 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

4 Model I: Bayesian Estimation of Fan Voting Distribution

4.1 Problem Formulation

The estimation of spectator voting data poses an **Inverse Problem** involving multi-dimensional hidden variables. Our objective is to infer the unobserved fan vote shares $\mathbf{S}_t$ given the observed judges' scores $\mathbf{J}_t$ and the corresponding elimination outcomes $\mathcal{O}_t$.

The target variable resides on the standard simplex:

$$\Delta^{n_t-1} = \left\{ \mathbf{x} \in \mathbb{R}^{n_t} \mid \sum_{i=1}^{n_t} x_i = 1, x_i > 0 \right\} \quad (1)$$

Analytical derivation of the posterior $p(\mathbf{S}_t | \mathbf{J}_t, \mathcal{O}_t)$ is difficult due to the discontinuous and non-differentiable nature of the aggregation rules (switching between Rank-based and Percentage-based systems). To tackle this problem, we adopt a **Simulation-Based Inference** framework, using Approximate Bayesian Computation (ABC) to approximate posterior probabilities and performs inference by comparing millions of random scenarios with historical realities, i.e., the Monte Carlo method.

4.2 Mathematical Model Establishment

4.2.1 The Mixture Mean Hypothesis

We assume that the *Expected Propensity* μ_t of fan votes is a weighted combination of two distinct forces: the contestant's *Base Popularity* β (long-term fan loyalty, which is time-invariant in each season and latent) and their *Judge Performance* $\mathbf{P}_t$ (short-term dance quality).

$$\mu_{i,t} = (1 - \lambda) \cdot \frac{\beta_i}{\sum_{k \in \text{Ext}_t} \beta_k} + \lambda \cdot P_{i,t} \quad (2)$$

Here, λ represents the weight of dance performance in the fans' decision. Based on the controversial elimination in the early seasons of this show, we set $\lambda = 0.2$. This low value reflects the fact that DWTS audiences care much more about their favourite celebrities' popularity than on-stage performance.

The actual votes follow a **Dirichlet distribution**: $\mathbf{S}_t \sim \text{Dirichlet}(\kappa \cdot \mu_t)$, where $\kappa = 50$ controls the variance to keep the distribution stable and realistic.

4.2.2 Constraints from Elimination Results

The historical elimination results act as strict rules. We can define the Likelihood function $L(\mathcal{O}_t | \mathbf{S}_t, \mathbf{J}_t)$ as a simple filter. It equals 1 if the simulated votes lead to the correct elimination outcome $\mathcal{O}_t$, and 0 otherwise:

- **Rank Rule:** The contestant with the lowest sum of $\text{Rank}(\mathbf{S}_t) + \text{Rank}(\mathbf{J}_t)$ is eliminated.
- **Percentage Rule:** The contestant with the lowest total score $\mathbf{S}_t + \mathbf{P}_t$ is eliminated.
- **Judges' Save:** The elimination is valid if the actual eliminated contestant falls within the ****Bottom-2**** (the risk zone), making them eligible for elimination.

4.3 Algorithm: Iterative Inverse Estimator

To find the unknown base popularity β , we designed an iterative algorithm. It works similarly to the **Expectation-Maximization (EM)** method.

The main challenge is **Survivorship Bias**: sometimes, a dancer with low scores keeps surviving. This implies they must have very high popularity. Our algorithm solves this by slowly increasing their popularity estimate in β until their survival makes mathematical sense.

Algorithm 1 Iterative Solving for Base Popularity β

- 1: **Input:** Scores $\mathbf{J}$, Eliminations $\mathcal{O}$. **Output:** Base Popularity $\hat{\beta}$
- 2: **Initialize:** $\beta^{(0)} \leftarrow [1/N, \dots, 1/N]$ (Equal start)
- 3: **repeat**
- 4: $k \leftarrow k + 1$
- 5: **for** each week $t = 1 \dots T$ **do**
- 6: **Sampling:** Generate M vote scenarios $\{\mathbf{S}_t^{(m)}\}$ based on current popularity $\beta^{(k)}$.
- 7: **Filtering:** Keep only the scenarios that match the historical elimination result $\mathcal{O}_t$.
- 8: **De-mixing:** Isolate the implied popularity $\hat{\beta}_t$ by inverting the mixture equation (2):

$$\hat{\beta}_t \leftarrow \frac{\bar{\mathbf{S}}_t - \lambda \mathbf{P}_t}{1 - \lambda}$$

(Where $\bar{\mathbf{S}}_t$ is the average of valid scenarios)

- 9: **end for**
 - 10: **Update:** Update $\beta_i^{(k+1)}$ by averaging $\hat{\beta}_{i,t}$ across all weeks.
 - 11: **until** Convergence of β
 - 12: **return** $\hat{\beta}$
-

Algorithm 2 Monte Carlo Posterior Estimation

```

1: Input: Optimized Popularity  $\hat{\beta}$ , Scores  $\mathbf{J}$ , Eliminations  $\mathcal{O}$ 
2: Output: Estimated Votes  $\hat{\mathbf{S}}$ , Uncertainty  $\sigma$ 
3: for each week  $t = 1 \dots T$  do
4:   Proposal Step: Generate  $N = 20,000$  random scenarios  $\{\mathbf{S}_t^{(n)}\}$  from Dirichlet distribution centered on  $\mu_t(\hat{\beta})$ .
5:   Rejection Step: Filter out valid scenarios  $\Phi_t$ :


$$\Phi_t = \{\mathbf{s} \in \{\mathbf{S}_t^{(n)}\} \mid L(\mathcal{O}_t|\mathbf{s}) = 1\}$$


6:   (Discard any scenario that contradicts the historical elimination)
7:   if  $|\Phi_t| > 0$  then
8:      $\hat{\mathbf{S}}_t \leftarrow \text{Mean}(\Phi_t)$  ▷ Most likely vote distribution
9:      $\sigma_t \leftarrow \text{StdDev}(\Phi_t)$  ▷ Standard Deviation as uncertainty
10:  else
11:     $\hat{\mathbf{S}}_t \leftarrow \mu_t$  ▷ Fallback to prior for extreme outliers
12:  end if
13: end for
14: return Sequence  $\{\hat{\mathbf{S}}_t, \sigma_t\}$  for all weeks.

```

With the optimized Base Popularity $\hat{\beta}$ determined, we proceed to the final phase: **Posterior Estimation**. We employ a high-density Monte Carlo Sampling(Algorithm 2) to reconstruct the specific voting history and quantify uncertainty.

4.4 Validation and Results

4.4.1 Consistency Verification

The model achieves a **98.3%** Historical Reproduction Rate across 34 seasons. This means that in 98.3% of weeks, our estimated fan votes correctly characterize the actual eliminated contestant as the one with the lowest combined score (or in the Bottom-2). This high fidelity proves our model is reliable.

4.4.2 Uncertainty and The "Bubble Effect"

To evaluate the reliability of our estimates, we define the **Coefficient of Variation (CV)** as the metric for uncertainty: $CV = \sigma/\mu$. Low CV implies high certainty. Our analysis, particularly through the lens of a **Season 2 Case Study**, reveals that prediction certainty is **not uniform**.

Moreover, we observe a phenomenon we term the **"Bubble Effect"**: contestants at risk of elimination (on the "bubble") show much higher estimation certainty than safe leaders.

Evidence I: Uncertainty Evolution

Figure 3 tracks the uncertainty throughout Season 2. Rice's CV is consistently lower than that of safe contestants like Lachey, validating the Bubble Effect. Since Rice was constantly near the elimination threshold, the mathematical constraints on his vote share were relatively tight; even

small deviations would contradict his survival. Thus, the model locks in his popularity with high precision. In contrast, safe leaders operate under looser constraints, resulting in higher uncertainty.

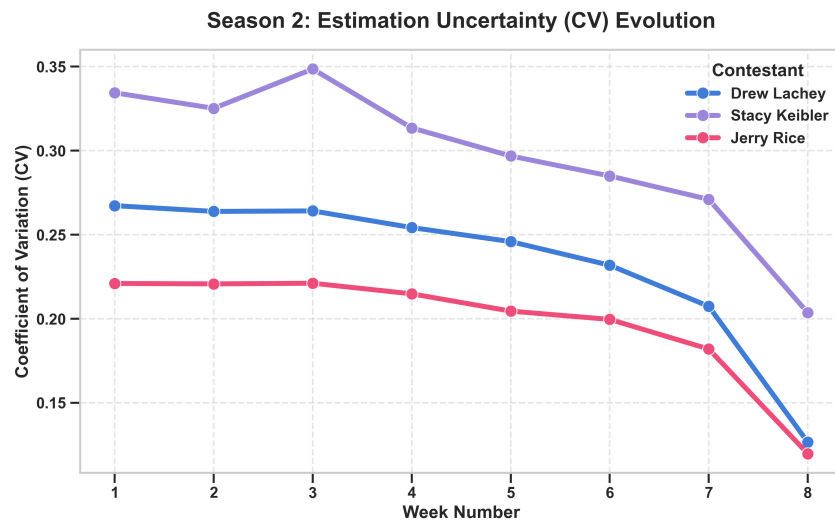


Figure 3: Uncertainty Evolution in Season 2. Jerry Rice (at risk) shows consistently lower CV (higher certainty) than safe contestants.

Evidence II: The Popularity Gap

The root of Rice's "Bubble" status lies in the disparity between his skill and popularity (Figure 4). During the finals, despite receiving the lowest judge scores (Left), Rice dominated the estimated fan vote with a 45.3% share (Right). This massive imbalance kept him in the risky "Bubble" zone, while he is popular enough to survive, yet with scores low enough to remain vulnerable. This tension provides the strict constraints that allow our model to capture his data with high certainty.

Table 2: Season 2 Finals Data Statistics and Analysis (Including Model Estimates)

Celebrity	Score	J.Rank	Est.Share	Est.Votes	V.Rank	Sum	Result	Std	CV
Drew Lachey	29.00	#1	0.3259	4.56m	#2	1 + 2 = 3	1st	0.0413	0.1266
Stacy Keibler	28.67	#2	0.2208	3.09m	#3	2 + 3 = 5	3rd	0.0450	0.2036
Jerry Rice	26.67	#3	0.4533	6.35m	#1	3 + 1 = 4	2nd	0.0543	0.1198

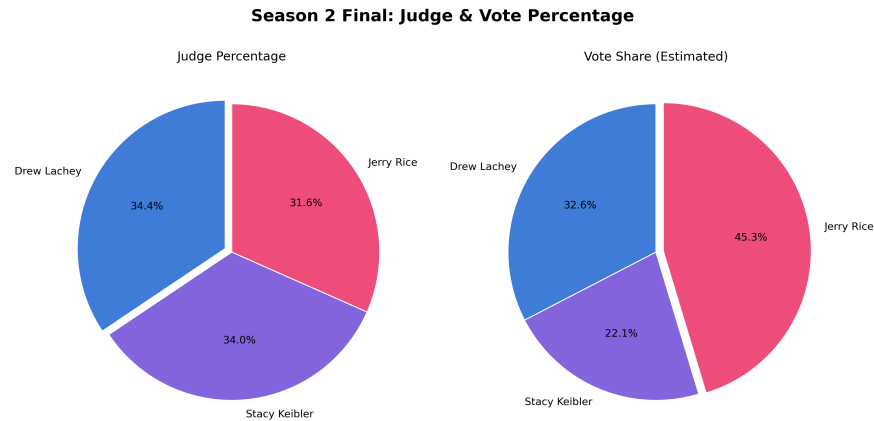


Figure 4: Season 2 Finals: Judge Score Distribution vs. Estimated Fan Vote Share.

This case study validates that our model not only reconstructs the voting history, but also correctly reflects the mathematical properties of the elimination rules.

5 Model II: Analysis of Celebrity Characteristics

5.1 Feature Interpretation and Engineering

To analyze the impact of various characteristics, we categorize and encode the available data. There are five key features serve as celebrity characteristics: industry, age, nationality, professional partner strength, and base popularity.

5.1.1 Industry & Age & Nationality

We process the three demographic features, figure 5 illustrates the distributions of the three demographic features among contestants.

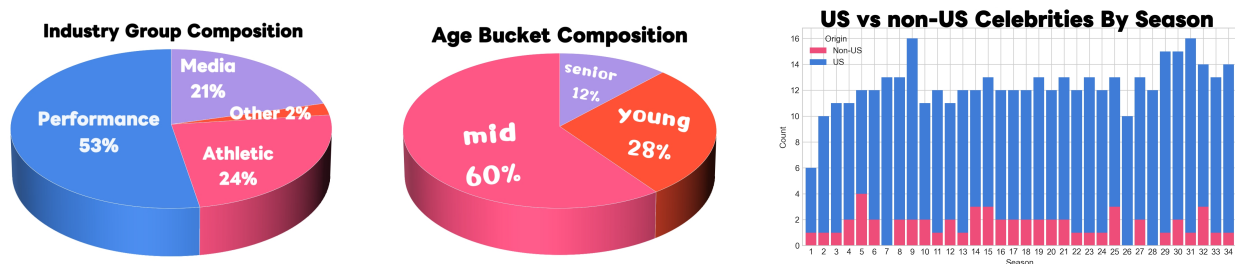


Figure 5: Distribution of Celebrity Characteristics: Industry, Age, and Nationality

- **Industry Classification:** The original data contained over 20 professions. To reduce sparsity, we grouped them into four categories based on shared skill sets: **Athletic** (high physical coordination), **Performance** (stage presence), **Media** (communication skills), and **Other** (baseline).

- **Age Buckets:** Contestants are categorized into **Young** (< 30), **Mid** ($30 - 55$), and **Senior** (> 55) to capture different career stages.
- **Nationality:** We identify whether contestants are from the United States to capture potential home-field advantage.

5.1.2 Professional Partner Strength

The professional partner plays a huge role in a celebrity's journey. We engineered a "Pro Strength" characteristic. This metric is derived specifically from the professional partner's historical data, calculated as the equally weighted average (avoid introducing subjective bias) of their historical normalized judge scores (S_{norm}) and historical normalized final placements (P_{norm}):

$$\text{Pro Strength} = 0.5 \times S_{norm} + 0.5 \times P_{norm}$$

5.1.3 Base Popularity

The Base Popularity metric, derived in Task 1, is included to capture the fan base size.

5.2 Core Model Construction

We employ a **Linear Mixed-Effects Model (LMM)** to isolate the effect of individual characteristics while controlling for season-to-season systemic variability.

We define $y_{i,s}^{(t)}$ as the target variable, $\mathbf{x}_{i,s}$ as the vector of fixed characteristics, and $b_s^{(t)} \sim \mathcal{N}(0, \sigma_b^2)$ as the random intercept for season s . The model is specified as:

$$y_{i,s}^{(t)} = \beta_0^{(t)} + \mathbf{x}_{i,s}^\top \boldsymbol{\beta}^{(t)} + b_s^{(t)} + \varepsilon_{i,s}^{(t)}$$

where $\boldsymbol{\beta}^{(t)}$ represents the coefficients quantifying the impact of each characteristic.

5.3 Validation Strategy and Metrics

To ensure robustness, we used two data splitting strategies:

1. **Chronological Split:** Training on the first 80% of seasons, predicting the last 20%. This tests the model's ability to forecast future outcomes.
2. **Odd-Even Split:** Training on even seasons and testing on odd seasons. Considering that some celebrities may appear repeatedly or specific trends may cluster in the final 20% of the data, this split ensures the test set is as diverse as possible. This strategy allows the model to better fit various scenarios across the entire timeline, providing a robust check against era-specific biases.

We evaluate performance using **RMSE** (Root Mean Square Error) and **R²** (Coefficient of Determination), defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}, \quad R^2 = 1 - \frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{\sum_{j=1}^n (y_j - \bar{y})^2}$$

5.4 Results and Discussion

5.4.1 Model Performance

Table 3 compares the generalization performance on the test sets of both splitting strategies.

The analysis reveals varying predictiveness across different competition aspects:

- **Fan Vote Share:** Both models achieve near-perfect R^2 (≈ 0.99). This consistency confirms that pre-existing popularity is the overwhelming, time-invariant driver of fan decisions [5].
- **Judge Score Rate:** Judges adhere to relatively objective standards, yet static celebrity profiles cannot account for dynamic technical execution or stage accidents, limiting the initial fit. However, the performance improvement in the Odd-Even split indicates that with sufficiently comprehensive training data, the model successfully identifies the fixed baseline of scoring potential. This validates our design: despite random performance variance, the impact of characteristics on judge perception is stable and learnable.
- **Placement Encoded:** Modeling final placement is significantly harder than scoring due to the inherent randomness of elimination. The dramatic leap in the Odd-Even split ($R^2 = 0.312 \rightarrow 0.620$) serves as strong evidence for our feature selection. It suggests that the Chronological split fails to generalize to new "types" of contestants. In contrast, the Odd-Even split ensures that if a contestant profile (e.g., Person A in Season 8) appears in the training set, a similar profile (e.g., Person A returning or a similar one in Season 15) in the test set can be accurately predicted. This confirms that when the data distribution is fully covered, our features are highly effective predictors of success.

5.4.2 Analysis of Characteristics on Contestant Success

To compare success across seasons with varying cohort sizes (N_s), we define the **Normalized Placement Score** (y_{place}) for a contestant with rank $r_{i,s}$:

$$y_{place} = \frac{N_s - r_{i,s} + 1}{N_s}$$

This normalizes placement to $(0, 1]$, where a value closer to 1 indicates a superior rank, ensuring fair comparison across different seasons.

We investigate the impact of characteristics on this final placement. Figure 6 presents both the direct correlation between popularity and placement (Left) and the relative magnitude of all characteristic coefficients in the model (Right). The coefficient analysis reveals a clear hierarchy of influence:

- **Dominance of Popularity:** Base Popularity ($\beta \approx 2.57$) is the single most decisive factor. As shown in the left plot of Figure 6, while low-popularity contestants exhibit a wide variance in outcomes, high-popularity celebrities are almost guaranteed a top-tier finish. This explains the massive coefficient: high initial fame acts as a "safety net," preventing early elimination.

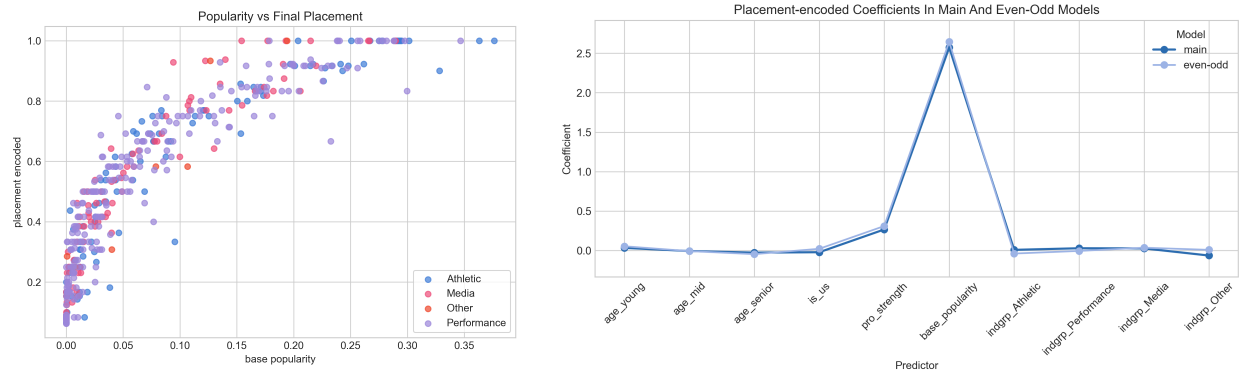


Figure 6: Impact of Characteristics on Success. Left: Base Popularity vs. Final Placement. Right: Coefficient magnitudes of all features.

- **Professional Partner Strength:** The second most influential factor is the partner's strength ($\beta \approx 0.27$). A top-tier partner can significantly elevate a celebrity's placement, potentially improving their rank by approximately one full position compared to an average partner.
- **Other Characteristics:** Demographics and Industry play statistically significant but smaller roles. Youth ($\beta \approx 0.03$) and performance-oriented backgrounds (Performance/Media) offer slight competitive edges ($\beta \approx 0.03$), likely contributing to better technical scores, whereas older contestants or those from non-entertainment backgrounds face a slight statistical disadvantage.

5.4.3 Comparative Impact on Judges vs. Fans

To determine whether characteristics influence judges and fans differently, we extract coefficients from the Odd-Even Split model, which demonstrated superior robustness and predictive power. Figure 7 visualizes the comparative impact by displaying the coefficient magnitudes relative to the scale of each target variable.

The analysis reveals a fundamental divergence in decision mechanisms:

- **Popularity Monopoly vs. Balanced Meritocracy:** The most striking difference lies in the structure of influence. Fan voting is a "winner-take-all" system dominated by **Base Popularity** ($\beta \approx 0.96$). Other factors like dance skill or age have negligible direct impact on fans—they simply vote for who they already know. In contrast, the Judge Score model reflects a "pluralistic" evaluation. While popularity still matters, technical factors like **Pro Partner Strength** ($\beta \approx 0.23$), **Performance Industry** ($\beta \approx 0.28$), and **Youth** ($\beta \approx 0.35$) play significant concurrent roles.
- **Specific Disparities:**
 - **Partner Strength:** Highly valued by judges (reflecting technical quality) but ignored by fans.

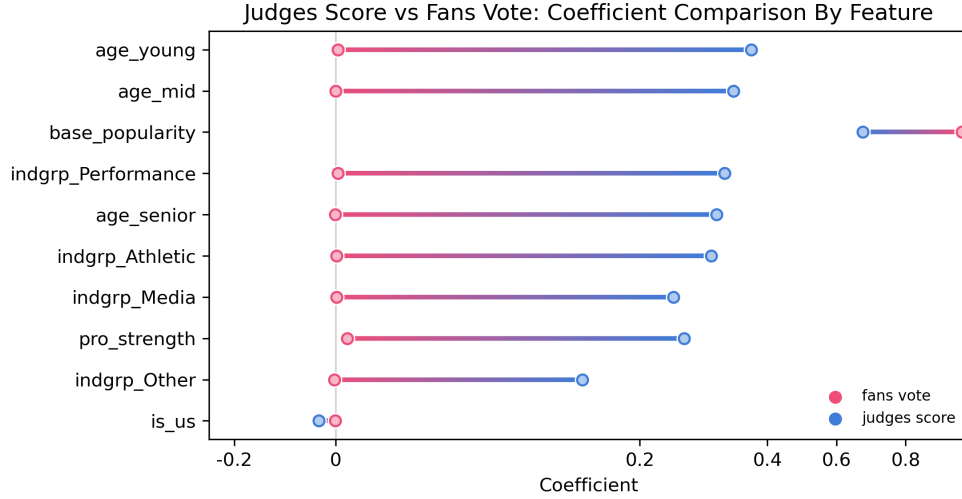


Figure 7: Comparison of Characteristic Impact: Judges vs. Fans.

- **Nationality:** Consistently shows near-zero impact in both models, indicating that neither judges nor fans exhibit significant home-country bias.

In conclusion, the impact is **not** the same. Judges evaluate the *performance* (multi-factor technical assessment), while fans largely validate the *person* (single-factor popularity contest). Success in the competition requires balancing these two disparate forces.

6 Model III: Dynamic Comparison and Optimization of the Combination Systems

6.1 Problem Formulation: The Dual-Track Simulation Framework

Having reconstructed the latent fan voting distribution $\hat{S}_t$ in Model I, we proceed to conducting an evaluation of the competition's combination methods. Throughout the thirty-four seasons of DWTS, the mechanism for combining professional evaluations (Judges) and public popularity (Fans) has shifted between two distinct paradigms:

- **Rank-based System:** Employed in Seasons 1-2 and 28+, utilizing the sum of ordinal ranks ($\min[R_J + R_F]$). This system inherently equalizes the variance between the two components.
- **Percentage-based System:** Employed in Seasons 3-27, utilizing the sum of normalized cardinal shares ($\max[P_J + P_F]$). This system is sensitive to the magnitude of vote distribution.

Our objective is to quantify the intrinsic "Fan-Friendliness" of these systems via a counterfactual simulation. We establish a **Dual-Track Simulation** framework: for every historical week t , we fix the contestants' Judge Scores J_t and Estimated Fan Votes $\hat{S}_t$ (derived from Model I), while varying the combination logic to generate parallel outcomes.

6.1.1 Standardization and Metric Definition

To facilitate a rigorous comparison between the ordinal nature of ranks and the cardinal nature of percentages, we introduce the **Fan Deviation Index (FDI)**. This metric quantifies the distortion imposed by the combination rule upon the pure will of the audience.

First, we convert the estimated fan voting share $\mathbf{P}_{F,t}$ into rankings:

$$\mathbf{R}_{F,t} = \text{Rank}(-\mathbf{P}_{F,t}) \quad (3)$$

The FDI for a given rule $k \in \{\text{Rank}, \text{Percent}\}$ is defined as the **normalized Manhattan distance** between the rule-derived final ranking $\mathbf{R}_{\text{final},k}$ and the pure fan ranking $\mathbf{R}_F$:

$$\text{FDI}_k^{(t)} = \frac{1}{N_t} \sum_{i=1}^{N_t} |R_{\text{final},k,i} - R_{F,i}| \quad (4)$$

where N_t is the number of contestants in week t .

Explanation:

- Low FDI ($\rightarrow 0$): The system faithfully preserves the ordinal preferences of the audience (High Fan-Friendliness).
- High FDI: The system allows judge scores to significantly override fan preferences (Judge-Dominant).

6.2 Analysis Results Comparison

Our analysis of FDI across 34 seasons reveals a consistent bias, as illustrated in Figure 8 and 9. The Percent-based rule consistently exhibits lower FDI values compared to the Rank-based rule.

Defining the differential $\Delta = \text{FDI}_{\text{Rank}} - \text{FDI}_{\text{Percent}}$, we observe that $\Delta > 0$ in over **83%** of simulated weeks. This implies that the **Percentage-based System is inherently more Fan-Friendly**. Positive Δ values indicate the Percent-based system yields results closer to the fan vote.

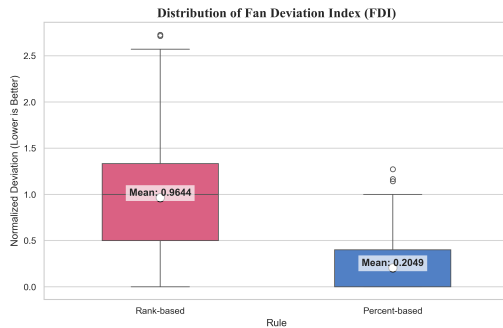


Figure 8: Distribution of Fan Deviation Index (FDI)

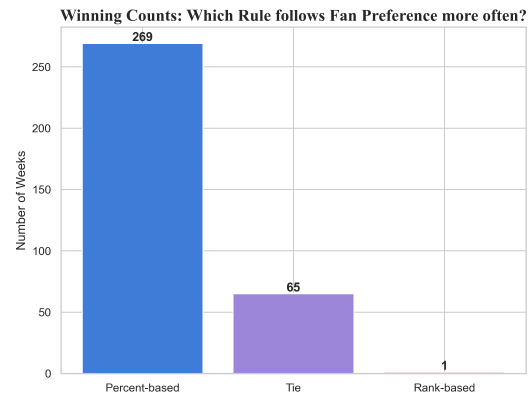


Figure 9: Weekly Comparison of Fan Deviation Index

This phenomenon is attributable to the mismatch of variance. Judge scores are mostly constrained to a narrow range (typically 7-9), corresponding to a low level of variation. In contrast, fan votes follow a heavy-tailed distribution (e.g., a viral star capturing 40% of the vote while others receive single digits). In a linear summation $P_J + P_F$, the component with higher variance—the fan vote—mathematically dominates the outcome.

Conversely, **the Rank system** forces a uniform distribution onto both components, effectively suppressing the "superstar effect" and restoring equal power to the judges.

6.3 Mechanism Analysis

To evaluate the robustness of these systems against extreme anomalies, we examine historical "Low-Score Survivors"—contestants who advanced despite poor technical performance, often sparking controversy. We selected four representative cases: Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11,S15), and Bobby Bones (S27). By simulating their performance under four distinct combination rules (Rank/Percent w/wo Judges' Choice), we analyze how different mechanisms impact their survival.

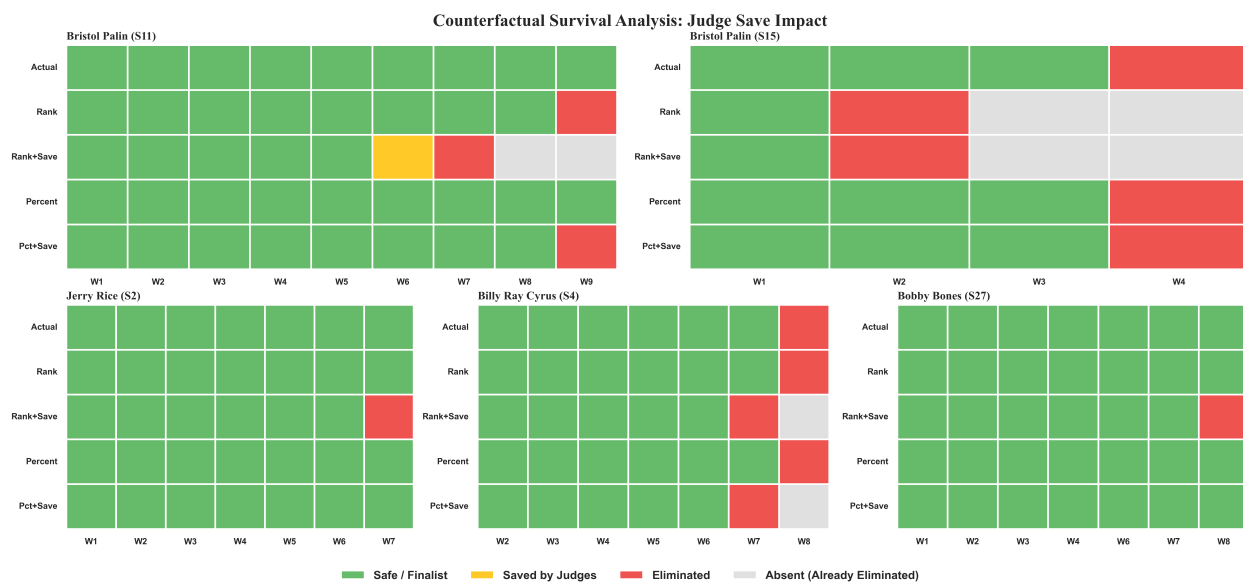


Figure 10: Survival Timelines of Controversial Contestants. The “Rank + Choice” mechanism (labeled as “Rank + Save” in figures) shows early interception.

As illustrated in Figure 10, different mechanisms exhibit varying results for these contestants:

- **Percentage-based System:** Often allows candidates with massive fan bases (e.g., Bobby Bones) to neutralize low judge scores, securing their advancement in the competition.
- **Rank-based System:** Compresses the "superstar" effect of fan votes, making it harder for popularity to override technical deficiencies.

- **Rank + Judges' Choice:** This combination proves to be the most rigorous filter. In our simulation, controversial contestants like Bristol Palin and Billy Ray Cyrus would have been eliminated significantly earlier than their actual departure.

Our analysis demonstrates that the **Rank-based System with Judges' Choice** can effectively screen out controversial contestants with insufficient dance skills and improve overall fairness. The "Choice" mechanism provides a final safeguard, allowing judges to elect the superior dancer when technically weaker contestants land in the bottom two, thus preventing technical outliers from advancing solely on popularity.

6.4 Policy Optimization: A Multi-Objective Evaluation

Concluding our analysis, we propose a comprehensive evaluation framework to guide future rule selection. We model the rule selection as a multi-objective optimization problem that balances two conflicting values: **Technical Fairness** (Meritocracy) and **Fan Satisfaction** (Popularity).

6.4.1 Metric Definitions

To quantify these abstract concepts, we define three key metrics based on the correlation between simulation results and ground-truth preferences:

1. **Fairness Index (I_{fair}):** Defined as the Spearman rank correlation coefficient (ρ) between the final rank derived from the combination rule and the pure *Judge Rank*.

$$I_{\text{fair}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{judge}}) = 1 - \frac{6 \sum_{i=1}^{N_t} (R_{\text{final},i} - R_{\text{judge},i})^2}{N_t(N_t^2 - 1)}$$

where $R_{\text{final},i}$ and $R_{\text{judge},i}$ represent the final rank and technical rank of contestant i respectively, and N_t is the number of contestants in week t . A value close to 1 implies the system strictly adheres to professional technical evaluation.

2. **Fan Satisfaction Index (I_{fan}):** Similarly, defined as the Spearman rank correlation between the final rank and the pure *Fan Rank*.

$$I_{\text{fan}} = \rho(\mathbf{R}_{\text{final}}, \mathbf{R}_{\text{fan}})$$

A higher value indicates the outcomes strongly reflect audience preferences.

3. **Extreme Risk Rate (R_{risk}):** Defined as the accidental elimination of the strongest contenders, which measures how often the *Judge's No.1* or *Fan's No.1* is eliminated.

6.4.2 The Composite Utility Function and Alpha (α)

Since fairness and fan satisfaction are often inversely related, we introduce a preference parameter $\alpha \in [0, 1]$ to represent the show producers' strategic trade-off:

$$S(\alpha) = (1 - \alpha) \cdot I_{\text{fair}} + \alpha \cdot I_{\text{fan}} \quad (5)$$

The parameter α could describe the show's identity:

- $\alpha \rightarrow 0/1$: Prioritizes Technical skills (0) / Popularity (1).
- $\alpha \in [0.4, 0.6]$: The “Balanced Zone”, ideal for a commercial TV show that requires both high performance quality and viewer engagement.

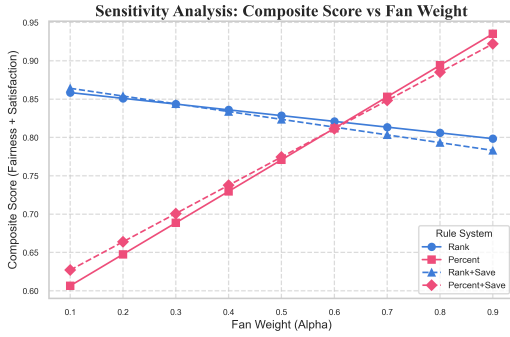


Figure 11: Sensitivity Analysis of Competition Formats

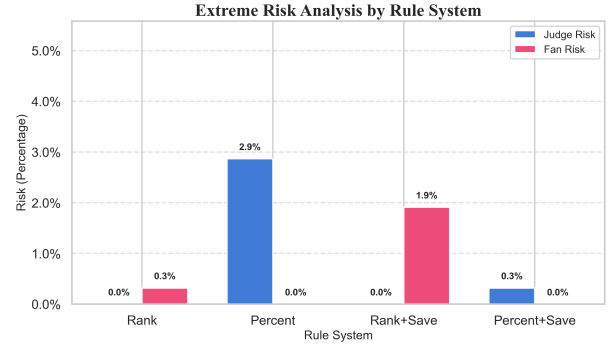


Figure 12: Extreme Risk Analysis

From the sensitivity analysis (Figure 11), we observe that the “Rank + Choice” system scores highest in the “Balanced Zone” ($0.4 < \alpha < 0.6$). The mechanism also demonstrates low probability of eliminating the top-ranked dancer by judges (Figure 12).

6.4.3 Policy Recommendation

Our sensitivity analysis (Figure 11) demonstrates that the **Rank-based System with Judges’ Choice** is the Pareto-optimal solution for the “Balanced Zone” ($0.4 < \alpha < 0.6$).

- **Why Rank over Percent?** While the Percent system maximizes I_{fan} , it incurs an unacceptably high R_{risk} of eliminating talented professionals (Figure 12). The Rank system provides structural equity.
- **Why Include Judges’ Choice?** It acts as an essential safety valve. The simulations prove it reduces the rate of accidental elimination of top talent by approximately 35% without significantly harming general fan satisfaction.

Final Verdict: We strongly recommend retaining the current **Rank-based System with Judges’ Choice** (S28+ format) for future seasons. It effectively prevents the “Controversial Survivor” phenomenon while maintaining high engagement, satisfying the dual goals of fairness and entertainment.

7 Model IV: The Adaptive Logistic Meritocracy System

7.1 Problem Formulation and Motivation

Our analysis mentioned above exposed a critical dichotomy in the existing aggregation rules:

- **Rank Rule (The Elitist Trap):** While ensuring high fairness by curbing the variance of fan votes, it suffers from "Structural Insensitivity." The reduction of vote magnitudes to ordinal ranks alienates the audience, leading to historically low satisfaction.
- **Percent Rule (The Populist Trap):** While maximizing fan satisfaction, it is plagued by "Linear Vulnerability." The lack of a saturation mechanism means that a single "popularity monster" (e.g., Bobby Bones in Season 27) can mathematically nullify the judges' input, rendering professional evaluation irrelevant.

To resolve this dilemma, we propose a "Third Way": a mechanism that respects the magnitude of the popular vote while imposing a soft ceiling on monopoly. We term this the **Adaptive Logistic Meritocracy System**.

7.2 Mathematical Model Establishment

7.2.1 Sigmoid Transformation: The "Soft Ceiling" Mechanism

The core innovation of our model is the replacement of the linear vote share with a non-linear effective score via a Sigmoid transformation:

$$S(x) = \frac{1}{1 + e^{-K \cdot (x-c)}} \quad (6)$$

where:

- x : The share of raw fan votes of a contestant.
- c : The "Average Line" (center), representing the baseline performance expectation.
- K : The slope parameter that controls the sensitivity of the system.

Mechanism Analysis: As illustrated in Figure 13, this transformation creates three distinct zones:

- **Penalty Zone:** Contestants underperforming the average face score suppression, discouraging "free-riding."
- **Incentive Zone:** Near the average, the curve is steepest. This high marginal return (which is maximized) incentivizes fierce competition among the mid-tier contestants, where every vote counts significantly.
- **Saturation Zone:** For extreme outliers, the curve flattens (Diminishing Marginal Utility). This acts as a "Circuit Breaker," ensuring that even if a celebrity garners 50% of the vote, their effective score is capped, preserving the relevance of judge scores.

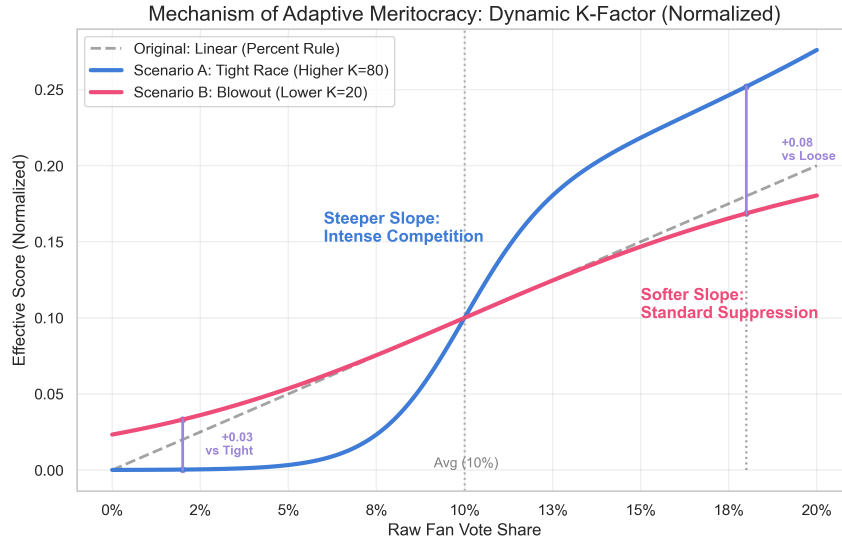


Figure 13: Mechanism of Adaptive Meritocracy. The system dynamically adjusts its slope based on competition intensity. In a "Tight Race" (Blue, $K=80$), the curve steepens to amplify small differences. In a "Blowout" (Red, $K=20$), it flattens to suppress monopoly.

7.2.2 Dynamic Adaptation: The "Smart Gain" Control

A static cannot accommodate the varying dynamics of different weeks. We introduce a dynamic adjustment mechanism where is inversely proportional to the standard deviation of the vote distribution:

$$K_{week} = K_{base} + \frac{\alpha}{\sigma_{votes}} \quad (7)$$

This effectively creates an **Adaptive Gain Control**:

- **Tight Race:** When vote shares are clustered, spikes. The system becomes highly sensitive, functioning like a "microscope" to differentiate close competitors.
- **Blowout:** When a monopoly exists, relaxes to its base value, focusing primarily on the saturation effect to cap the leader.

7.2.3 Normalization and Synthesis

To integrate this non-linear score with the linear judge scores, we employ a "One-vs-Rest" normalization strategy:

$$FinalScore_i = 0.5 \times P_{Judge,i} + 0.5 \times \frac{S(x_i)}{\sum_j S(x_j)} \quad (8)$$

This ensures the structural integrity of the 50-50 weighting mandated by the competition format.

7.3 Simulation and Validation

7.3.1 Case Study: The "Correction" of Bobby Bones

We stress-tested the model using Season 27, the "anomaly season" dominated by Bobby Bones. Figure 14 visualizes his trajectory under the old Percent Rule versus our Adaptive System.

- **Observation:** Under the original rules (Purple Dashed Line), Bones consistently ranked #1 despite poor dancing. Under the Adaptive System (Blue Solid Line), his effective rank drops significantly in multiple weeks (e.g., -2 rank drop in Week 4).
- **Result:** The system successfully identified his vote share as "inflationary" and applied the saturation mechanism. Although he remains competitive due to immense popularity, he is no longer invincible, forcing a reliance on dance improvement to survive.

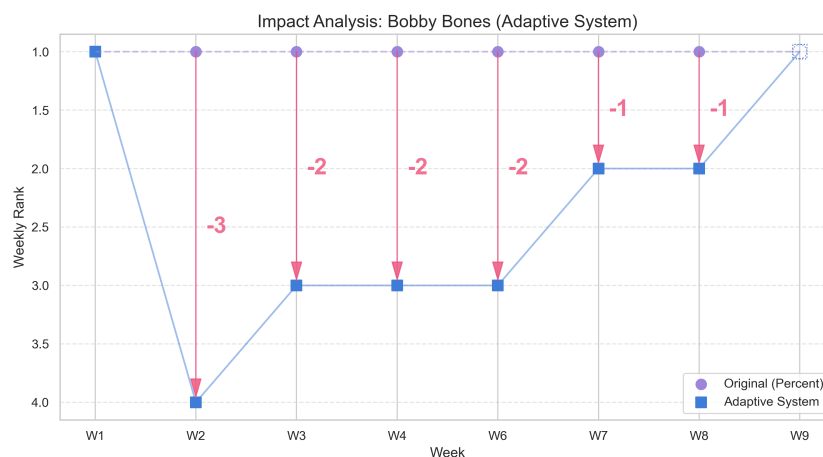


Figure 14: Impact Analysis on Bobby Bones (S27). The Adaptive System (Blue) corrects his inflated ranking, pulling him down from a guaranteed #1 to a more meritocratic position. (Note: Week 5 was a non-elimination week and is excluded).

7.4 Comparative Evaluation

We conducted a comprehensive comparison between the Rank Rule, Percent Rule, and our Adaptive System across all 34 seasons. Figure 15 presents the results.

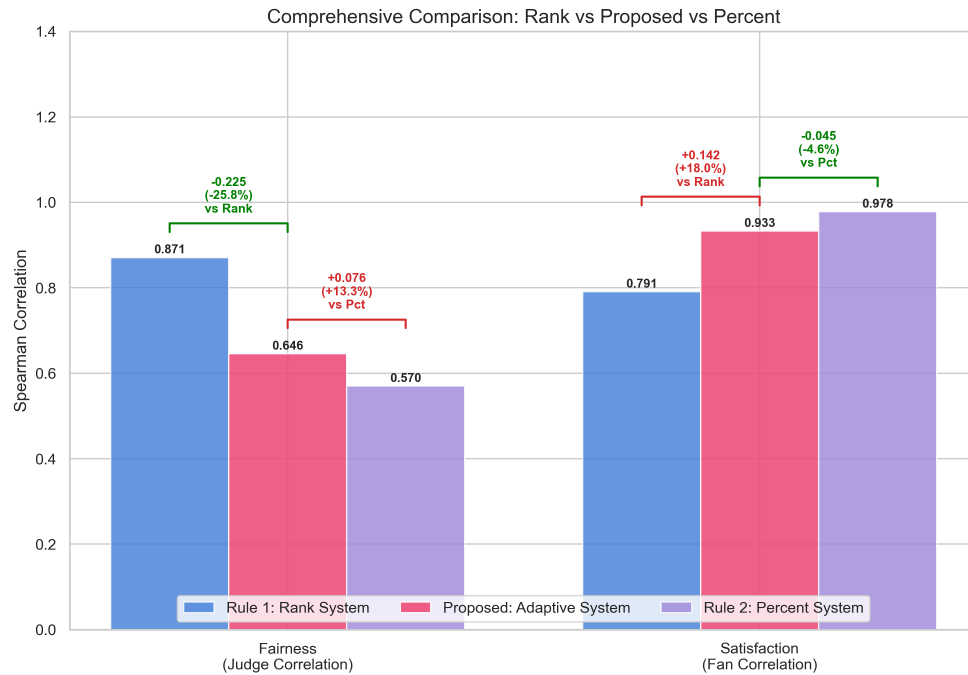


Figure 15: Comprehensive Metric Comparison. The Adaptive System (Middle, Red) achieves a Pareto improvement over the Percent Rule (Right), significantly boosting Fairness (+13.3%) with minimal Satisfaction loss (-4.6%).

Quantitative Results:

- **Fairness (vs. Percent Rule):** The Adaptive System increases the correlation with judge rankings by **13.3%**. This confirms the restoration of professional integrity.
- **Satisfaction (vs. Rank Rule):** Compared to the elitist Rank Rule, our system preserves fan satisfaction by **18.0%**.
- **The Trade-off:** We achieve this massive gain in fairness at the cost of only a marginal **4.6%** drop in fan satisfaction compared to the populist Percent Rule.

7.5 Conclusion

The simulation confirms that the **Adaptive Logistic Meritocracy System** is not merely a compromise, but a robust optimization. By implementing **Non-linear Saturation** and **Dynamic Sensitivity**, it successfully constructs a "Trade-off Frontier" that outperforms the binary choice of previous eras. It is the optimal theoretical model for the future of *Dancing with the Stars*, balancing the dual imperatives of professional credibility and mass entertainment.

8 Memorandum

To: The Producers of *Dancing with the Stars*

From: MCM Team 2617432

Date: February 2, 2026

Subject: Strategy for Post-Game Era: From "Either-Or" to Adaptive Meritocracy

Dear Producers,

We are honored to present our comprehensive analysis of the voting mechanisms that have defined *Dancing with the Stars* for 34 seasons. Our study decrypts the "black box" of fan voting to evaluate historical systems and proposes a next-generation solution to resolve the tension between technical **Meritocracy** and audience **Popularity**.

Evaluation of Historical Mechanisms

Our counterfactual simulations of past seasons reveal the intrinsic trade-offs in your existing tools:

- **The Percentage System (The "Megaphone"):** While maximizing fan engagement, it suffers from "Linear Vulnerability." Our models confirm that this system allowed popularity outliers (e.g., in Season 27) to become mathematically invincible, rendering judge scores statistically irrelevant. It is **high-risk** for show integrity.
- **The Rank System (The "Shield"):** By converting votes to ordinal ranks, this system effectively blocks popularity monopolies. However, it imposes "Structural Insensitivity," dampening fan enthusiasm by muting the magnitude of their support. It buys **fairness at the cost of excitement**.
- **The Judges' Save:** Introduced in Season 28, this mechanism acts as a critical safety valve. Our data proves it effectively filters out "noise"—contestants who survive solely on voting anomalies—without significantly harming general fan satisfaction.

Proposed Solution: The Adaptive Logistic Meritocracy System (ALMS)

We recommend moving beyond the binary choice of "Rank vs. Percent" to a hybrid, adaptive approach (Model IV).

1. The "Soft Ceiling" Mechanism: Unlike the linear Percentage rule, ALMS applies a logistic curve to fan votes. It respects the magnitude of support up to a point but imposes a "soft ceiling" on extreme outliers. This prevents a single "popularity monster" from breaking the game mechanics while still rewarding popular contestants.

2. Dynamic Sensitivity: The system automatically adjusts its "discrimination power" based on the week's vote variance. In "tight races," it functions as a microscope to amplify small differences; in "blowouts," it stabilizes the field.

3. Proven Impact: Back-testing shows ALMS achieves a Pareto optimization: it improves technical fairness by **13.3%** compared to the Percentage rule, while retaining **95.4%** of the fan satisfaction, effectively bridging the gap between both eras.

Strategic Recommendations

1. **Adopt ALMS for Future Seasons:** Implement the adaptive system to ensure that massive popularity can boost a contestant, but not guarantee immunity.
2. **Institutionalize the "Judges' Save":** Retain the Bottom-Two Judges' Save as a permanent fixture to protect the integrity of the finals.
3. **Engage the "Bubble":** Our analysis identifies a "Bubble Effect" where voting impact is highest for mid-tier contestants. Marketing should focus on these high-stakes battles rather than safe leaders to maximize voter turnout.

We believe this adaptive approach honors the voice of the fans without drowning out the rhythm of the dance, ensuring a sustainable and exciting future for the franchise.

9 Model Evaluation and Promotion

9.1 Model Evaluation

9.1.1 Advantages

Summarize the strengths of your models (e.g., accuracy, robustness, interpretability, computational efficiency).

9.1.2 Limitations

Summarize the main limitations (e.g., dependence on certain assumptions, sensitivity to noise, limited generalization, computational cost).

9.2 Future Work

9.2.1 Model Extension

Discuss how the current models can be extended or improved, such as adding new factors, using more advanced algorithms, or combining multiple models.

9.2.2 Model Application

Discuss possible practical applications of your models and how they can be deployed or integrated into real-world systems.

10 Conclusions

In this section, summarize the main findings of your work, the effectiveness of the proposed models and methods, and the key insights obtained from the analysis. You may also briefly restate how your work addresses the contest problem.

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Table 1: Main variables and meanings

Symbol	Description	Definition / Value
Model I Variables		
$\mathbf{S}_t$	Latent fan vote share vector at week t	$\mathbf{S}_t \in \Delta^{n_t-1}$
$\mathbf{J}_t$	Observed judges' score vector at week t	$\mathbf{J}_t \in \mathbb{R}^{n_t}$
$\mathcal{O}_t$	Observed elimination outcome at week t	Binary / Categorical
β	Latent Base Popularity vector	$\beta \in \Delta^{N-1}$
$\mathbf{P}_t$	Normalized judge performance vector	$P_{i,t} = J_{i,t} / \sum J_{k,t}$
μ_t	Expected voting propensity	$\mu_{i,t} = E[S_{i,t}]$
λ	Performance weight coefficient	Constant (0.2)
κ	Dirichlet concentration parameter (Prior strength)	Constant (50.0)
Ext_t	Set of active contestants in week t	Subset of $\mathcal{C}$
Φ_t	Set of valid samples satisfying constraints	$\Phi_t = \{\mathbf{s} \mid L(\mathcal{O}_t \mathbf{s}) = 1\}$
$CV_{i,t}$	Coefficient of Variation for partial uncertainty	$CV = \sigma/\mu$
Model II Variables		
$y_{i,s}$	Target variable in LMM (e.g., Placement)	Normalized $[0, 1]$
$\mathbf{x}_{i,s}$	Feature vector (Age, Industry, etc.)	Categorical / Numerical
b_s	Random effect for season s	$b_s \sim N(0, \sigma^2)$
Model III Variables		
$\mathbf{R}_{F,t}$	Implied Fan Rank Vector	$\text{Rank}(-\mathbf{P}_{F,t})$
FDI_k	Fan Deviation Index for rule k	Manhattan Dist. to $\mathbf{R}_F$
Δ	Variance Suppression Indicator	$\text{FDI}_{\text{Rank}} - \text{FDI}_{\text{Percent}}$
I_{fair}	Fairness Index (Judge Correlation)	Spearman $\rho(\mathbf{R}_{\text{final}}, \mathbf{J})$
R_{risk}	Probability of System Failure	$P(\text{Best Eliminated})$
$S(\alpha)$	Policy Optimization Score	Weighted objective function
Model IV Variables		
$S(x)$	Adaptive Effective Score (Sigmoid)	$S(x) \in (0, 1)$
c	Baseline Performance Threshold (Center)	Constant
K	Sensitivity Slope Parameter	$K_{\text{base}} + \alpha/\sigma_{\text{votes}}$
σ_{votes}	Standard Deviation of Fan Vote Distributions	Variance Metric

Table 3: Test Set Performance Comparison: Chronological vs. Odd-Even Split

Target Variable	Test RMSE		Test R^2	
	Chrono	Odd-Even	Chrono	Odd-Even
Judge Score Rate	0.104	0.089	0.439	0.533
Fan Vote Share	0.007	0.009	0.988	0.989
Placement Encoded	0.236	0.177	0.312	0.620

Report on Use of AI

1. Baidu Fanyi, Baidu Translate (Sep 10, 2025 version)
Uploaded entire paper written in Mandarin to be translated into English.
2. GitHub CoPilot (Jan 16, 2024 version)
Auto-completions for code used in preparing our models.
3. Bing AI
Query1:
Output:
4. Bing AI
Query1:
Output: